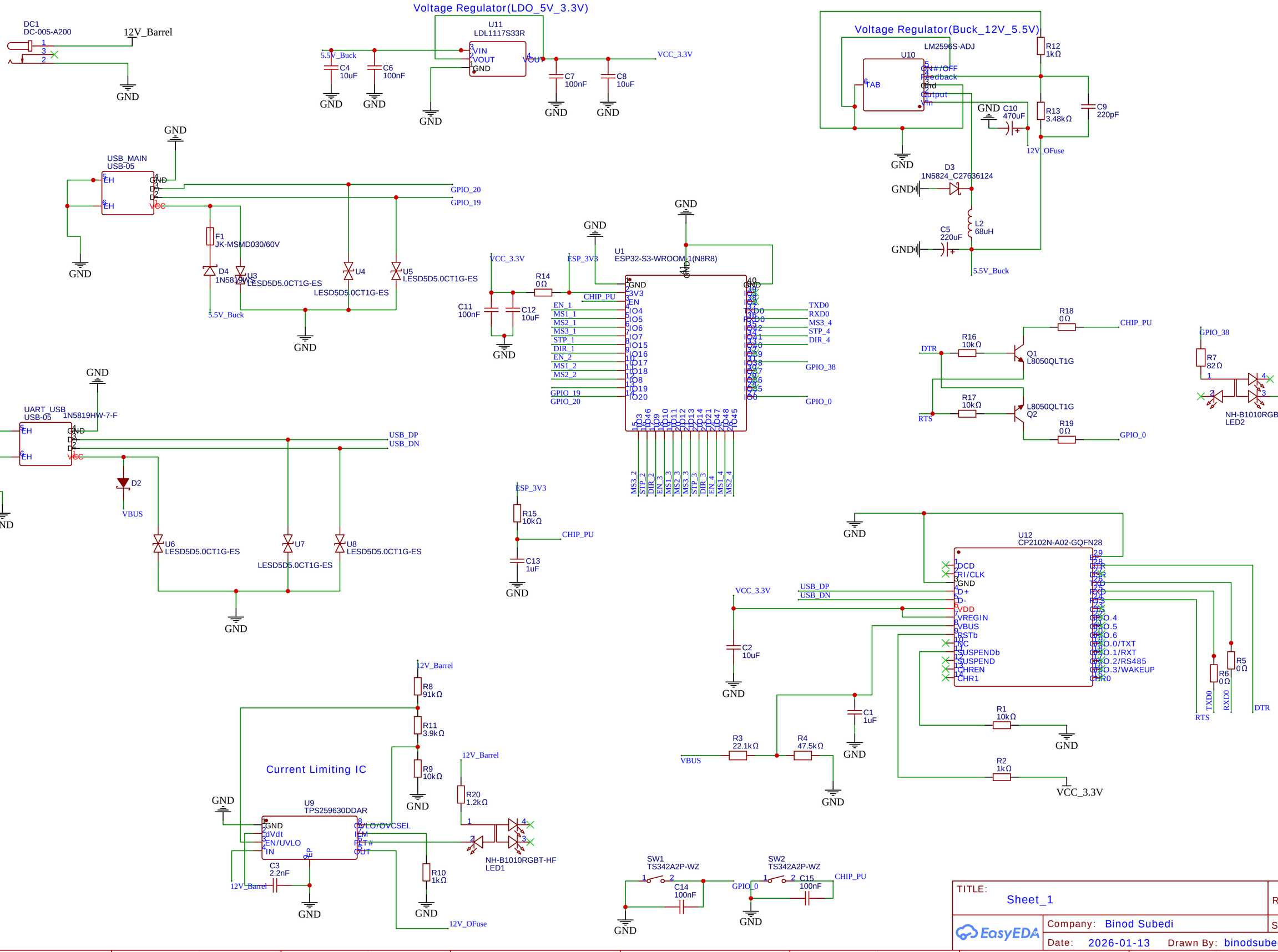
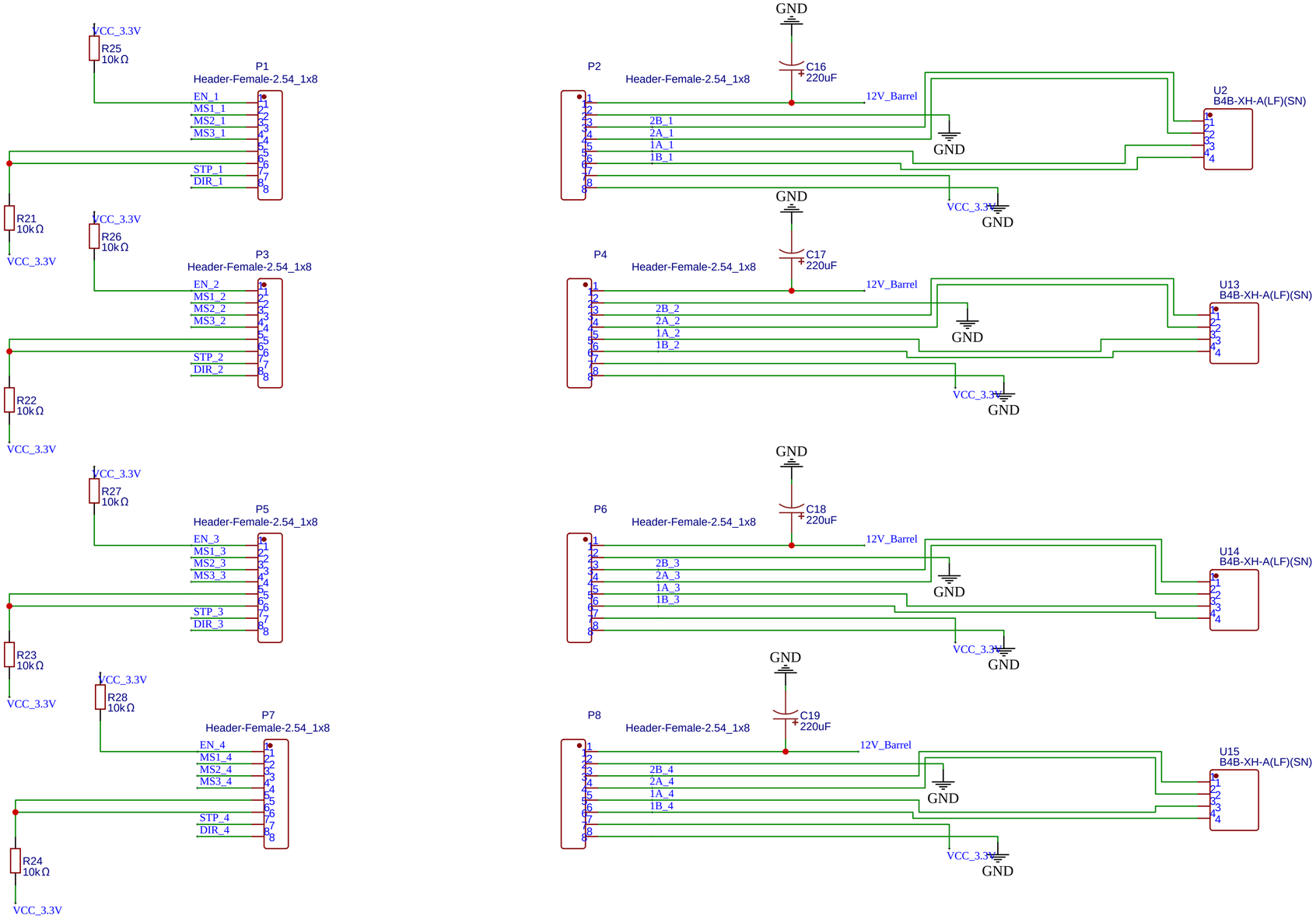

Voltage Regulator(LDO_5V_3.3V)
DC1
DC-005-A200
12V_Barrel
GND
U11
LDL1117S33R
VIN
VOUT
GND
5.5V_Buck
C4
10uF
C6
100nF
C7
100nF
C8
10uF
VCC_3.3V
Voltage Regulator(Buck_12V_5.5V)
LM2596S-ADJ
U10
ON#/OFF
Feedback
Gnd
Output
Vin
TAB
R12
1kΩ
R13
3.48kΩ
C9
220pF
C10
470uF
12V_OFuse
D3
1N5824_C27636124
C5
220uF
L2
68uH
5.5V_Buck
USB_MAIN
USB-05
EH
GND
D+
D-
VCC
GPIO_20
GPIO_19
F1
JK-MSMD030/60V
D4
1N5819WS
U3
LESD5D5.0CT1G-ES
U4
LESD5D5.0CT1G-ES
U5
LESD5D5.0CT1G-ES
U1
ESP32-S3-WROOM-1(N8R8)
VCC_3.3V
ESP_3V3
R14
0Ω
C11
100nF
C12
10uF
CHIP_PU
EN_1
MS1_1
MS2_1
MS3_1
STP_1
DIR_1
EN_2
MS1_2
MS2_2
GPIO_19
GPIO_20
TXD0
RXD0
MS3_4
STP_4
DIR_4
GPIO_38
GPIO_0
R18
0Ω
R16
10kΩ
DTR
Q1
L8050QLT1G
R17
10kΩ
RTS
Q2
L8050QLT1G
R19
0Ω
GPIO_38
R7
82Ω
NH-B1010RGBT-HF
LED2
UART_USB
USB-05
1N5819HW-7-F
USB_DP
USB_DN
D2
VBUS
U6
LESD5D5.0CT1G-ES
U7
LESD5D5.0CT1G-ES
U8
LESD5D5.0CT1G-ES
ESP_3V3
R15
10kΩ
CHIP_PU
C13
1uF
U12
CP2102N-A02-GQFN28
DCD
RI/CLK
D+
D-
VDD
VREGIN
VBUS
RSTb
NC
SUSPENDb
SUSPEND
CHREN
CHR1
GPIO.4
GPIO.5
GPIO.6
GPIO.0/TXT
GPIO.1/RXT
GPIO.2/RS485
GPIO.3/WAKEUP
CHR0
C2
10uF
C1
1uF
R1
10kΩ
R2
1kΩ
R3
22.1kΩ
R4
47.5kΩ
R5
0Ω
R6
0Ω
TXD0
RXD0
RTS
DTR
VCC_3.3V
Current Limiting IC
U9
TPS259630DDAR
GND
dVdt
EN/UVLO
IN
OVLO/OVCSEL
ILIM
RETRY#
OUT
EP
C3
2.2nF
R8
91kΩ
R11
3.9kΩ
R9
10kΩ
R20
1.2kΩ
R10
1kΩ
NH-B1010RGBT-HF
LED1
12V_Barrel
12V_OFuse
SW1
TS342A2P-WZ
SW2
TS342A2P-WZ
C14
100nF
C15
100nF
GPIO_0
CHIP_PU
TITLE:
Sheet_1
REV: 1.0
Company: Binod Subedi
Sheet: 1/2
Date: 2026-01-13
Drawn By: binodsubedi125
EasyEDA

Stepper Motor drivers pins



Note: SLP and RST needs to get pulled high with 10k resistor, also need to connect the VMOT with decoupling capacitor of at least 100uf.

Note: Also will pull the EN pin down with 10k resistor, to have the default mode to enable.

TITLE: Sheet_2		REV: 1.0
EasyEDA	Company: Binod Subedi	Sheet: 2/2
	Date: 2026-01-21 Drawn By: binodsubedi125	